```
import pandas as pd
import numpy as np
```

```
df = pd.read_csv('/content/customer_shopping_behavior.csv')
```

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Di: A
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day	

```
df.tail()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type
3895	3896	40	Female	Hoodie	Clothing	28	Virginia	L	Turquoise	Summer	4.2	No	2-Day Shipping
3896	3897	52	Female	Backpack	Accessories	49	Iowa	L	White	Spring	4.5	No	Store Pickup
3897	3898	46	Female	Belt	Accessories	33	New Jersey	L	Green	Spring	2.9	No	Standard
3898	3899	44	Female	Shoes	Footwear	77	Minnesota	S	Brown	Summer	3.8	No	Express

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df=df.rename(columns={'purchase_amount_(usd)' : 'purchase_amount'})
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
```

```
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

```
df.isnull().sum()
```

	0
customer_id	0
age	0
gender	0
item_purchased	0
category	0
purchase_amount	0
location	0
size	0
color	0
season	0
review_rating	37
subscription_status	0
shipping_type	0
discount_applied	0
promo_code_used	0
previous_purchases	0
payment_method	0
frequency_of_purchases	0

dtype: int64

```
df['review_rating'] = df.groupby('category')['review_rating'].transform(lambda x : x.fillna(x.median()))
```

```
df.isnull().sum()
```

```
labels = ['young','adult','middle_aged','senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

```
df[['age','age_group']].head()
```

	age	age_group
0	55	middle_aged
1	19	young
2	50	middle_aged
3	21	young
4	45	middle_aged

```
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['frequency_purchase_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

```
df[['frequency_purchase_days','frequency_of_purchases']].head()
```

	frequency_purchase_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually

```
df.to_csv("Consumer_Shopping_Behavior", index=False)
```

```
from google.colab import files
files.download("Consumer_Shopping_Behavior")
```